

Shen (Sam) Xie

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samxiexs



shenxie

EDUCATION

Georgia Institute of Technology, Atlanta, United States

Aug 2026 – Present

Master's, Major in Computational Science and Engineering; H. Milton Stewart School of Industrial and Systems Engineering
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Wuhan University, Wuhan, China

Sep 2020 – Jun 2024

Bachelor's, Major in Politics, Philosophy, and Economics; Minor in Computer Science and Engineering; Hongyi Honor College

- Wuhan University, Hongyi Honor College **Honor Graduate; First Prize Academic Scholarship (ranked 2nd)**.

RESEARCH INTERESTS

AI for Scientific Discovery, Multimodal Learning, Recommender Systems, Human-AI Interaction, Digital Platforms, UGC.

RESEARCH EXPERIENCE

HEC Paris, Paris, France

Mar 2025 – Aug 2026

Remote Research Assistant, Department of ISOM; supervisor: Prof. Xitong Li

- Conducting researchs related to ML and HCI.
- Built real-time data pipelines for **Reddit, SSRN, FastMoss, Tiktok Shop**, producing research datasets for creator-economy, platform consumer behavior, and digital marketing studies.
- Designed influencer-product matching workflows with rule-based and machine-learning approaches.

Hong Kong University of Science and Technology, Hong Kong SAR, China

Aug 2024 – Jul 2025

Onsite Research Assistant, Department of ISOM; supervisor: Prof. Tatkooh Koh

- Reviewed literature on crowdsourcing contests, human-AI interaction, diversification, and platform strategy to support theory building and empirical design. Built a large-scale **99designs** dataset.

Wuhan University, Wuhan, China

Feb 2024 – Jun 2024

Research Assistant, Department of MSE; supervisor: Prof. Baojun Gao

- Conducted robustness checks and **Structural Topic Modeling** for an **Journal of Retailing (ABS 4)** paper under second review using TripAdvisor review data.

ONGOING RESEARCH PAPERS

ML: General-Purpose LLM Agents for Efficient Scientific Experiment Selection

Sep 2026 – Present

Co-first author with Yuming Zeng; Preparing for submission to ICML 2027

- Developing a general framework for **sequential scientific experiment selection**, where an agent adaptively chooses the most informative next experiment from a large candidate space to accomplish a scientific objective with as few experiments as possible.

ML: Diffusion-Based Image–Audio Alignment

Aug 2026 – Present

Third Author with Biyang Xin, Mingwei Sun, Shijie Zhang, and Xitong Li; In progress

- Develops a diffusion-based multimodal framework for learning aligned **image–audio representations**, enabling cross-modal correspondence and generation across visual and acoustic modalities.

ML: EEG-to-Speech Decoding via Coordinate-Aware Cross-Modal Representation Learning

Apr 2026 – Present

First Author with Mingwei Sun, Shijie Zhang, and Xitong Li; Preparing for submission to ICML 2027

- Develops a unified **EEG-to-speech-content** framework that maps variable-channel EEG signals to shared **MFCC** and **HuBERT** speech representations using coordinate-aware CNN–Conformer encoders.
- Builds leakage-controlled multi-dataset pipelines with subject/content OOD splits, masked channel–time modeling, contrastive EEG–audio alignment, and counterfactual EEG controls for rigorous evaluation.

HCI: When VLM Frames the Product Advertising

Nov 2025 – Aug 2026

Second author. Under review at CHI 2027; Revising for Journal submission

- Details as well as the tiles withheld during peer review.

IS: The Power of Content Similarity in Recent Reviews on Hotel Revenue

May 2025 – Present

Second author with Baojun Gao and Jiawen Wang; In preparation to submit to

International Journal of Hospitality Management (ABS 3);  *GitHub*

- Analyzes **1.5M TripAdvisor reviews** and revenue data from **2,990 hotels** across Texas cities.
- Finds that higher adjacent review similarity significantly reduces revenue, especially for high-end, high-volume, and lower-rated hotels.

SELECTED GITHUB PROJECTS

Software Engineering: Influencer Agent for TikTok Shop

Sep 2025 – Aug 2026

with start-up: InFound.ai;  *GitHub*

- Led backend and experimentation development for an **AI-native influencer marketing system** for MACOS and Windows, covering creator search, profiling, automated outreach and chat, sample management, sales-data processing, and API / MySQL / service integration.
- Built real-time processing and deployment-facing infrastructure that connects creator matching, recommendation delivery, and operational analytics into an end-to-end data loop from outreach to fulfillment.
- Received the **Gold Award** at the Guangdong-Hong Kong-Macao Greater Bay Area Inno Awards Demo Day.

ML Engineering: Generative Ad Recommendation with RQ-VAE Semantic IDs and RankMixer Bidding

Oct 2025 – Mar 2026

with Xitong Li

- Learned **RQ-VAE**-based discrete semantic codes for joint image-text **5120-d embeddings**, compressing them into **4-d $\times$ 256-way Semantic IDs** with **over 92%** utilization in all 4 codebooks.  *GitHub*
- Designed a **RankMixer** bidding reranker with parameter-free token mixing, per-token FFN / **SparseMoE**, and joint **CTR+CTCVR** training, delivering offline **+0.5% AUC** and online lifts in participation, win, impression, and **PctCVR**.  *GitHub*

Web Crawling: Social Science Papers Auto Crawler and Analysis Agent

Apr 2025 – Jun 2025

with Ziqing Li;  *GitHub*

- Built a crawling agent and dataset covering more than **1.2 million papers**, including titles, abstracts, author information, and citation links.
- Designed the pipeline to support downstream topic discovery, citation analysis, and automated literature exploration.

SEMINAR OUTREACH

- Chinese Open Science Network Hackathon 2025
invited by Southern University of Science and Technology and University of Macau Aug 2025
- Quant Marketing PhD Summer Workshop
invited by City University of Hong Kong Jul 2025
- AI for Business Summer Workshop
invited by University of Sci and Tech of China; Outstanding Participant Award Jul 2025
- Paper Sharing: Unexpected Negative Effect from Platform Regulation
Management Sci Seminar co-hosted by Wuhan University and Xi'an JiaoTong University May 2025

SKILLS

- Programming languages: **Python, R, Stata, SQL, $\LaTeX$** .
- Engineering stack: **FastAPI, Redis, RabbitMQ, Docker, PostgreSQL, MySQL**.
- Research and modeling: **recommender systems**, NLP, structural topic modeling, multimodal modeling, and large-scale web crawling.